Steerable Microwave Beam Phased Array

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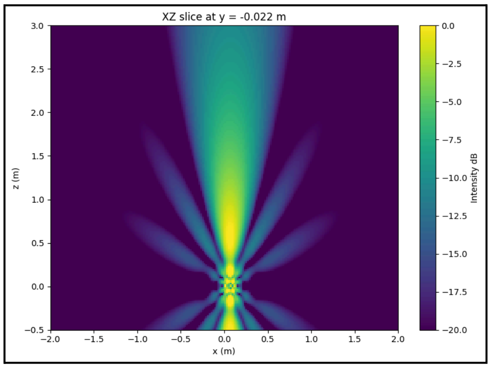
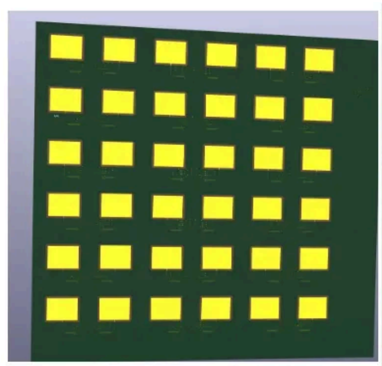
Document revision: 1.1
Status: In design





1. Project Overview

This project explores the design and construction of a steerable microwave phased array capable of delivering a tightly focused beam of microwave energy toward a target. This project will be my submission for the **OSHWLab Stars 2026** PCB design competition.

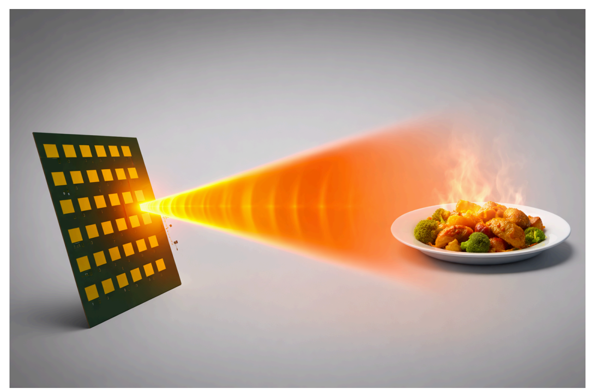
1.1 Purpose

The purpose of this project is to develop and **improve my technical skills** in areas that excite me, all while building a cool project that will **show flashy, mind-bending results**. The system serves as a science demonstration platform that reveals the power and elegance of microwave circuit design, phased arrays, and high-frequency PCB engineering. The core concept is to electronically steer RF energy without any moving mechanical parts, using phase control across an array of patch antennas.

1.2. Key Results

A successful project will have the following results:

- Delivers focused microwave energy across a distance
  - heats up food, illuminates LEDs across a room
- Focuses energy beam as tightly as possible
  - ~17° beam waist or 0.5m diameter at 2m
- Implementation of phased-array beamforming
  - Steers over a range of 75° in 1 axis
  - 4 programmable angles
- Extra things to show off (stretch goals)
  - Plasma generation inside noble gases
  - Recover radiated energy using a receiver antenna array (wireless drone charger?)

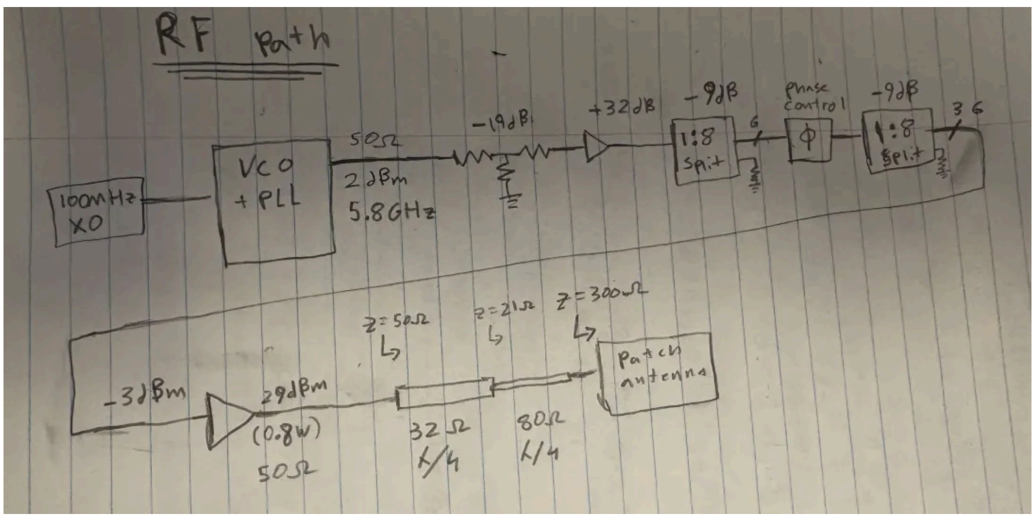


2. System Architecture

The beam is formed using a phased array of microstrip patch antennas. Each antenna element is driven with controlled phase offsets, allowing constructive interference in a desired direction and destructive interference elsewhere.

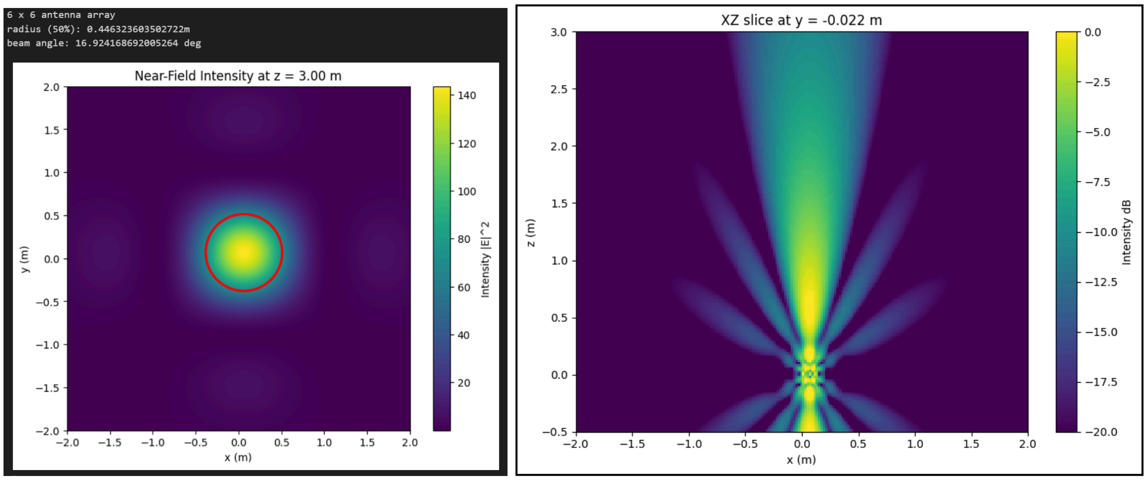
Core blocks include:

- RF signal generation
- Power amplification and attenuation
- RF splitting network
- Phase shifting network
- Impedance matching and length tuning
- Patch antenna design & array layout



3. Design Process

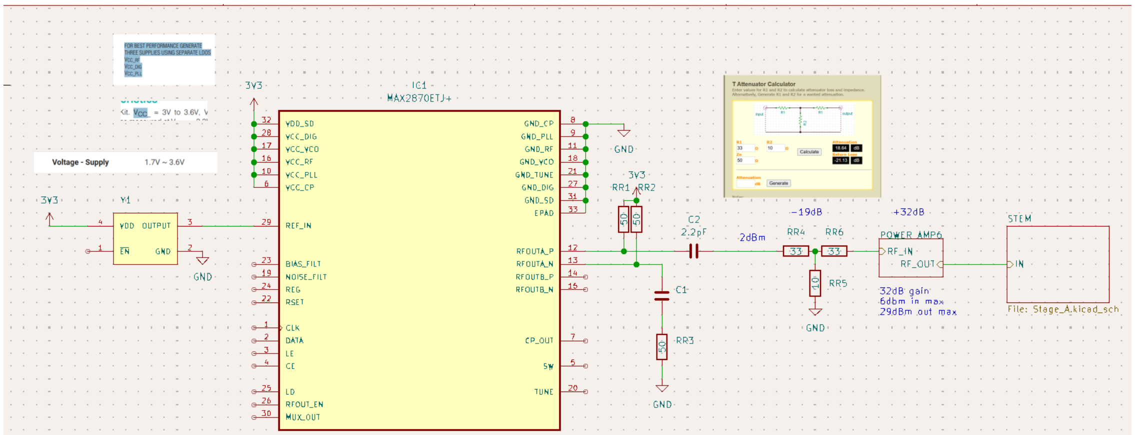
The design process involved iteration between simulation and physical layout. Electromagnetic simulations were used to validate antenna resonance, impedance matching, and beamforming characteristics.

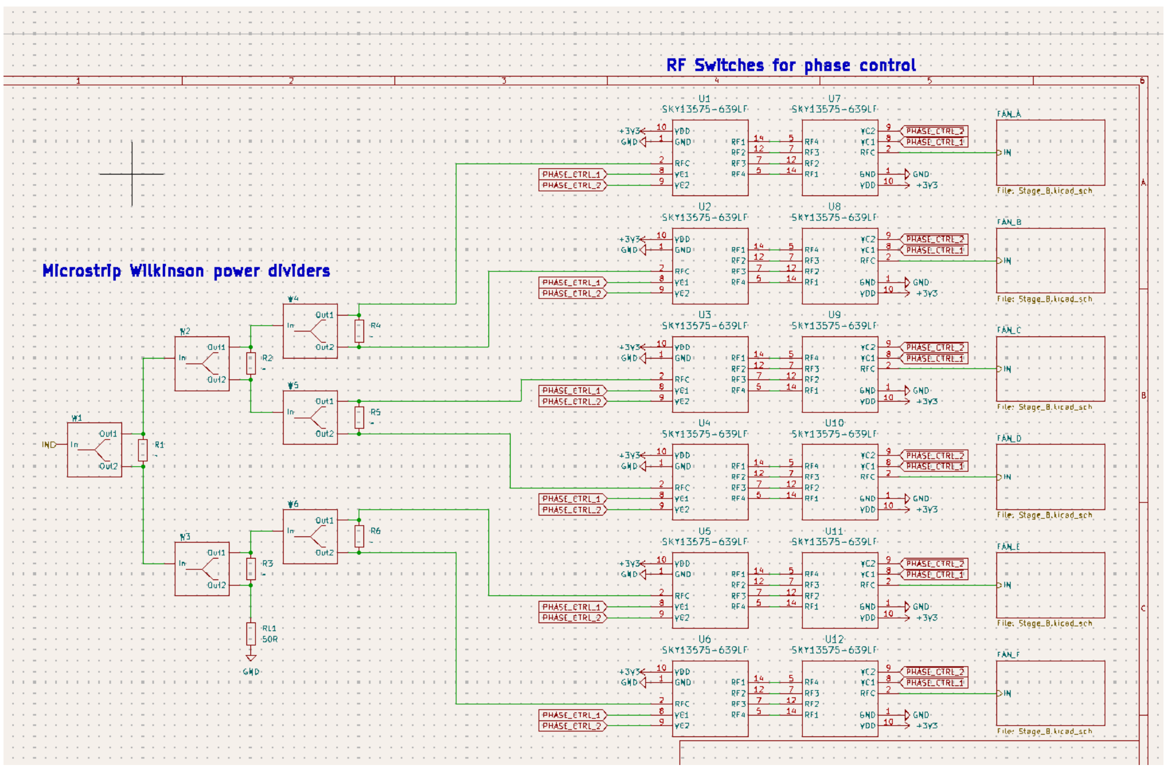


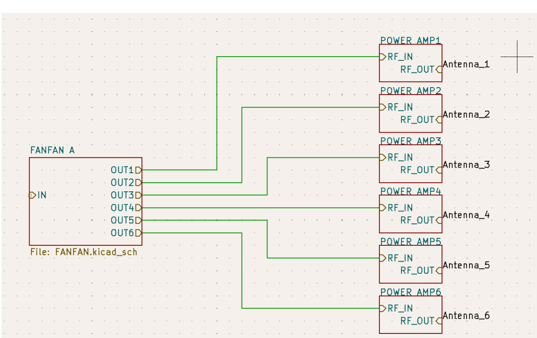
Simulating the beam from my array using python (simplifying the antennas into ideal point sources) reveals a 3dB beam dispersion of 17 degrees.

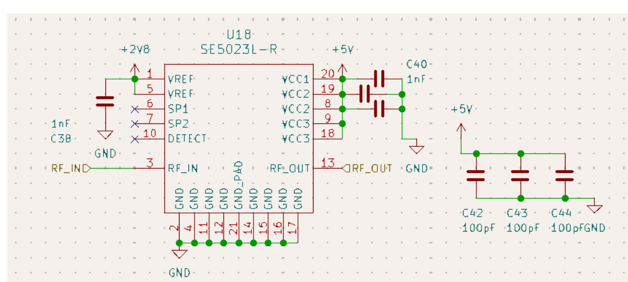
3.1. Schematic design

Hierarchical sheets are heavily used in the schematic due to the repetitive nature of the circuit.



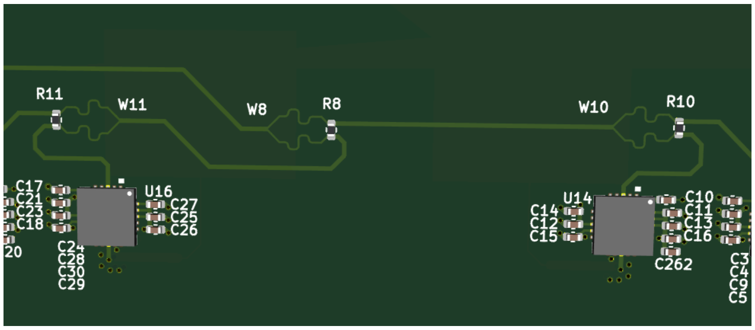




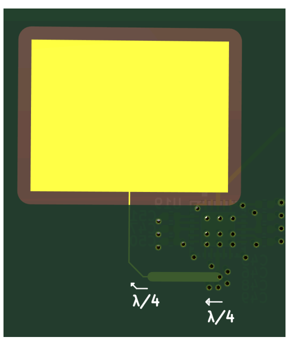


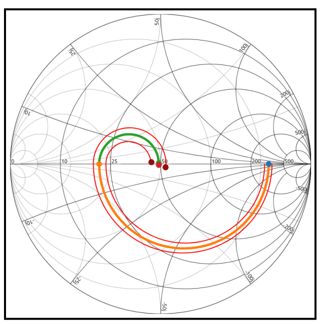
4. Microstrip Impedance Matching

I use **Wilkinson power dividers** to maintain matched impedance in the signal lines as well as provide isolation between the outputs.

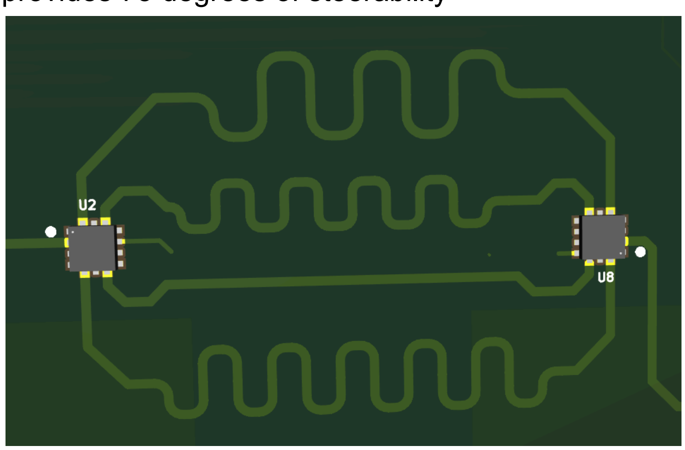


Quarter wavelength transformers are used to match the ~300 ohm patch antennas to the 50 ohm amplifiers. QWT was chosen over stub or lumped element matching due to its robustness: 20% inaccuracies in either the length or line impedance still lands with a manageable 10dB return loss.





Phase control is achieved using SP4T switches to select the appropriate delay path for the chosen beam steering angle. The chosen steering angles in degrees are: 45, 20, 0, -30. This provides 75 degrees of steerability



5. Technical Challenges

• Sourcing RF components that operate reliably at the target microwave frequency
• Managing PCB dielectric tolerances affecting antenna tuning & electrical length
• Maintaining consistent phase across RF paths
• Thermal considerations under higher power levels
• Avoiding unintended radiation or coupling between array elements

Most resistors and capacitors do not provide documentation of performance at high frequencies - especially as high as 5.8GHz. Below shows impedance vs frequency charts for resistors chosen for the T attenuator, and capacitors chosen for DC blockers. Murata's "sim surf" website was very handy for choosing appropriate capacitors.